Store Versile miniature room frame: manufacturing guide (Rev C)

Box: 300 × 200 × 25 mm inside · Printer: Bambu Lab A1 Mini (single colour, 0.4 mm nozzle) · PLA · Scale about 1:12

1. Status and what was checked

Check	Result
STL files	45 files (44 parts + optional LED channel), 63 printed pieces per frame
Watertight, one closed body, no non-manifold edges (trimesh)	**All pass**
Fits A1 Mini 180 × 180 × 180 in print orientation	**All pass** (largest: arch 78 × 176, wall panel 104 × 168, shelf unit 78 × 166)
Parts overlapping in the assembly (`cad/check_interference.py`)	**None**
Sliced with PrusaSlicer 2.7 (Bambu Studio is built on it) with an A1-Mini-like profile, **no supports**	**All slice**
Overhangs steeper than 45°	Only small rounded edges (largest 35 mm² on the coffee-table chamfer). Flat ceilings are short bridges of 0.3–4 mm (frame photo slot, panel key notches, book covers)
Bambu Studio slicing and a physical test print	**Still to do on your side.** Nothing here is called production-ready until the test prints in section 8 fit

Per frame: 63 pieces, about **212 g** of PLA, about **8 h 25 min** of printing if every part is printed alone. Batching many parts on one plate per colour is faster. These estimates come from PrusaSlicer; Bambu Studio's numbers will be close but not identical (expect ±30 %).

2. Bill of materials

Printed: see the print list (section 3). **Bought per frame:**

Item	Qty	Note
Shadow-box frame, inside 300 × 200 × 25 mm, white back	1	Your existing frame
Warm-white LED strip, 5 mm wide, about 300 mm, with battery box or USB	1	Under the top front edge (optional printed channel: `LED_Channel` × 2)
PLA filament off-cuts, 1.75 mm	6 × 6.4 mm	Shelf pins
Gel CA glue (super glue)	—	Plastic to plastic
Glue stick or photo-safe double-sided tape	—	Paper prints
Photo paper, max 0.3 mm thick	1 A4	Arch print, 8 photo inserts, quote
Sticker paper (or paper + glue)	1 A4	Wood floor, book spines
Pink acrylic paint, fine brush	—	Bunny inner ears (optional)

3. Print list

Settings that apply to every part: PLA, 0.4 mm nozzle, **supports OFF**, first layer 0.2 mm, nozzle 220 °C, bed 65 °C, gyroid infill (rectilinear at 100 %). Use Bambu Studio's *0.12mm Fine*, *0.16mm Optimal* or *0.20mm Standard* A1 Mini profile to match the layer column. Each STL already lies in its print orientation, so keep it as imported.

STL file	Qty	Colour	Size W×D×H mm (as printed)	Orientation	Layer	Infill	Walls	Brim	g each	Time each	Notes
Floor_Tile_L.stl	1	Any colour	110 × 25 × 2	As installed (bottom down)	0.2	15%	2	–	6.11	8m 55s	Wood sticker on top after printing.
Floor_Tile_C.stl	1	Any colour	112 × 25 × 2	As installed (bottom down)	0.2	15%	2	–	6.65	9m 40s	Wood sticker on top after printing.
Floor_Tile_R.stl	1	Any colour	86 × 25 × 2	As installed (bottom down)	0.2	15%	2	–	4.75	7m 5s	Wood sticker on top after printing.
Left_Wall_Panel_Lower.stl	1	Peach-beige	104 × 31.99 × 2.4	On its back, front up	0.16	15%	2	–	6.82	13m 19s	Flutes up. Seam is hidden behind the TV console.
Left_Wall_Panel_Upper.stl	1	Peach-beige	104 × 168 × 2.4	On its back, front up	0.16	15%	2	–	38.93	1h 0m 33s	Flutes up. 6 shelf-pin holes.
Right_Wall_Panel_Lower.stl	1	Peach-beige	78 × 31.99 × 2.4	On its back, front up	0.16	15%	2	–	5.12	10m 8s	Flutes up. Seam is hidden behind the shelving.
Right_Wall_Panel_Upper.stl	1	Peach-beige	78 × 168 × 2.4	On its back, front up	0.16	15%	2	–	29.20	43m 9s	Flutes up.
Floating_Shelf.stl	3	Ivory white	55 × 8 × 3	As installed (bottom down)	0.16	100%	3	–	1.60	3m 22s	Print 3. Two pin holes in the back edge.
LOVE_Letters.stl	1	White	22.47 × 17.83 × 4	On its back, front up	0.12	100%	2	–	1.21	5m 39s	Letters face up.
Photo_Frame_Small_P.stl	4	Gold silk	24 × 27 × 2.5	On its back, front up	0.12	100%	2	–	0.99	2m 45s	Gold face up (ironing on). Photo slot opens at the top.
Photo_Frame_Small_L.stl	3	Gold silk	27 × 24 × 2.5	On its back, front up	0.12	100%	2	–	0.99	2m 45s	Gold face up (ironing on). Photo slot opens at the top.
Photo_Frame_Large_P.stl	2	Gold silk	30 × 39 × 2.5	On its back, front up	0.12	100%	2	–	1.73	4m 1s	Gold face up (ironing on). Photo slot opens at the top.
TV_Console_Body.stl	1	Black	102 × 24 × 10	On its back, front up	0.16	15%	3	–	12.27	25m 14s	Lying on its back: bays and

STL file	Qty	Colour	Size W×D×H mm (as printed)	Orientation	Layer	Infill	Walls	Brim	g each	Time each	Notes
											drawer recess face up.
TV_Console_Drawer_Fronts.stl	1	White	96.7 × 9.2 × 0.8	On its back, front up	0.12	100%	2	–	0.89	2m 37s	0.8 mm plate, grooves up. Glue into the body recess.
TV_Console_Leg.stl	4	White	3.76 × 2.2 × 11.6	As installed (bottom down)	0.12	100%	2	3	0.05	51s	Print 4 together with a 3 mm brim, slow small-layer cooling on.
Gramophone_Base.stl	1	Dark brown	14.98 × 8.1 × 13.48	As installed (bottom down)	0.12	20%	2	–	1.19	8m 58s	Angled hole for the horn on top.
Gramophone_Horn.stl	1	Gold silk	11 × 7.92 × 14	Upside down	0.12	100%	2	–	0.09	2m 6s	Bell mouth down on the bed, peg up.
Book_01.stl	2	White	24 × 8 × 5.5	As installed (bottom down)	0.12	20%	2	–	0.74	3m 47s	Spine label = printed paper.
Book_02.stl	2	Black	22.5 × 7.5 × 5.5	As installed (bottom down)	0.12	20%	2	–	0.66	3m 40s	Spine label = printed paper.
Book_03.stl	1	White	23.5 × 8 × 5.5	As installed (bottom down)	0.12	20%	2	–	0.72	3m 46s	Spine label = printed paper.
Plant_Pot_01.stl	2	White	14.9 × 9.93 × 29	As installed (bottom down)	0.12	15%	3	–	3.46	22m 21s	Print 2. Two parallel leaf slots.
Plant_Leaves_01A.stl	2	Leaf green	33.44 × 29.32 × 1	On its back, front up	0.12	100%	2	–	0.42	5m 43s	Print 2. Flat, 1.0 mm. Front palm fan.
Plant_Leaves_01B.stl	2	Leaf green	33.54 × 30.26 × 1	On its back, front up	0.12	100%	2	–	0.34	4m 42s	Print 2. Flat, 1.0 mm. Rear palm fan.
Plant_Pot_02.stl	1	White	24 × 7 × 12	As installed (bottom down)	0.12	15%	3	–	1.92	9m 3s	
Plant_Leaves_02.stl	1	Leaf green	28.44 × 20.97 × 1	On its back, front up	0.12	100%	2	–	0.31	1m 19s	Flat, 1.0 mm.
Plant_Pot_03.stl	1	White	15 × 7 × 12	As installed (bottom down)	0.12	15%	3	–	1.09	8m 59s	Diamond-knurl texture.

STL file	Qty	Colour	Size W×D×H mm (as printed)	Orientation	Layer	Infill	Walls	Brim	g each	Time each	Notes
Plant_Leaves_03.stl	1	Light green	17.89 × 18.85 × 1.2	On its back, front up	0.12	100%	2	–	0.17	1m 2s	Flat, 1.2 mm.
Plant_Pot_04.stl	2	White	14 × 7 × 10	As installed (bottom down)	0.12	15%	3	–	0.86	7m 7s	Print 2 (shelf unit + TV console).
Plant_Leaves_04.stl	2	Leaf green	13.06 × 13 × 1	On its back, front up	0.12	100%	2	–	0.10	58s	Print 2. Flat, 1.0 mm.
Central_Arch.stl	1	White	78 × 176 × 3	On its back, front up	0.16	15%	2	–	17.93	26m 48s	Border up. 176 mm long: place diagonally if your plate edge is tight; no brim needed.
Name_Plate.stl	1	White	20 × 8 × 1.6	On its back, front up	0.12	100%	2	–	0.22	1m 12s	FILAMENT CHANGE at 1.0 mm: white base, then dark text. Edit the NAME line in this file per order.
Coffee_Table.stl	1	White	60 × 14 × 26	Upside down	0.12	20%	2	–	2.73	19m 16s	Upside down. FILAMENT CHANGE at 2.6 mm: white top first, then wood legs.
Serving_Tray.stl	1	Light wood	38 × 10 × 1.5	As installed (bottom down)	0.12	100%	2	–	0.69	2m 19s	Ripple grooves up.
Mug.stl	2	White	9.8 × 7.5 × 8	As installed (bottom down)	0.12	100%	2	–	0.20	2m 56s	Print 2 (one is turned so the handles face outward).
Arch_Vase.stl	1	Beige	15 × 14 × 5	On its back, front up	0.12	20%	2	–	0.71	3m 36s	Arch grooves up.
Silver_Heart_Plant.stl	1	Silver silk	12.68 × 11.73 × 2.11	On its back, front up	0.12	100%	2	–	0.17	1m 27s	Bumps up. Push the stem into the arch vase.
Right_Display_Shelf.stl	1	Cream	78.2 × 165.7 ×	On its back, front up	0.2	15%	3	–	39.93	55m 26s	Lying on its back:

STL file	Qty	Colour	Size W×D×H mm (as printed)	Orientation	Layer	Infill	Walls	Brim	g each	Time each	Notes
			12								every board is a vertical wall, no supports, no bridges.
Gold_Ornament.stl	1	Gold silk	9.97 × 13.68 × 4.98	On its back, front up	0.12	20%	2	–	0.41	3m 11s	Flat back down.
Gold_Ornament_Leaves.stl	1	Leaf green	14.01 × 11.2 × 1	On its back, front up	0.12	100%	2	–	0.07	39s	Glue the tab to the back of the gold pot.
Takeaway_Cup.stl	1	White	9.3 × 9.3 × 22	As installed (bottom down)	0.12	20%	2	–	1.43	13m 30s	Plain cup, no logo. Optional own-brand sticker on the sleeve.
Donut_Vase.stl	1	White	12 × 13.6 × 3	On its back, front up	0.12	20%	2	–	0.34	2m 17s	Flat back down.
Pampas_Spray.stl	1	Bone / ivory white	11.03 × 13.74 × 1	On its back, front up	0.12	100%	2	–	0.07	54s	Flat, 1.0 mm. Peg goes into the donut vase neck.
Bunny.stl	1	White	13.15 × 24.68 × 6.18	On its back, front up	0.12	20%	2	–	0.71	4m 39s	Flat back down. Paint inner ears pink.
Gold_Elephant.stl	1	Gold silk	16.65 × 14.82 × 5.2	On its back, front up	0.12	20%	2	–	0.70	4m 42s	Flat back down.
LED_Channel.stl	2	White	148 × 5 × 6	Upside down	0.2	15%	2	–	1.82	3m 40s	Optional, print 2. Hides the LED strip under the top front edge.

4. Plates by colour (batch printing)

Group parts by colour and fill each 180 × 180 plate. Small figurines print much better several at a time (each layer gets time to cool). Totals per frame:

Colour	Pieces	Filament	Time (parts printed alone)	Parts
Peach-beige (e.g. Bambu PLA Matte Desert Tan / Latte Brown, or PLA Sparkle for shimmer)	4	80.1 g	2 h 07 min	Left_Wall_Panel_Lower x1, Left_Wall_Panel_Upper x1, Right_Wall_Panel_Lower x1, Right_Wall_Panel_Upper x1

Colour	Pieces	Filament	Time (parts printed alone)	Parts
Cream (e.g. PLA Matte Ivory White; or the panel colour)	1	39.9 g	55 min	Right_Display_Shelf x1
White (e.g. PLA Basic Jade White / Matte Ivory White)	23	39.9 g	2 h 54 min	LOVE_Letters x1, TV_Console_Drawer_Fronts x1, TV_Console_Leg x4, Book_01 x2, Book_03 x1, Plant_Pot_01 x2, Plant_Pot_02 x1, Plant_Pot_03 x1, Plant_Pot_04 x2, Central_Arch x1, Name_Plate x1, Coffee_Table x1, Mug x2, Takeaway_Cup x1, Donut_Vase x1, Bunny x1
Any colour (covered by the wood-grain sticker)	3	17.5 g	26 min	Floor_Tile_L x1, Floor_Tile_C x1, Floor_Tile_R x1
Black (PLA Basic / Matte Charcoal)	3	13.6 g	33 min	TV_Console_Body x1, Book_02 x2
Gold silk (e.g. PLA Silk+ Gold)	12	11.6 g	38 min	Photo_Frame_Small_P x4, Photo_Frame_Small_L x3, Photo_Frame_Large_P x2, Gramophone_Horn x1, Gold_Ornament x1, Gold_Elephant x1
Ivory white (e.g. PLA Matte Ivory White)	3	4.8 g	10 min	Floating_Shelf x3
Leaf green (e.g. PLA Matte Grass Green)	8	2.1 g	25 min	Plant_Leaves_01A x2, Plant_Leaves_01B x2, Plant_Leaves_02 x1, Plant_Leaves_04 x2, Gold_Ornament_Leaves x1
Dark brown (e.g. PLA Matte Dark Brown)	1	1.2 g	9 min	Gramophone_Base x1
Beige (e.g. PLA Matte Desert Tan)	1	0.7 g	4 min	Arch_Vase x1
Light wood (e.g. PLA Matte Latte Brown or PLA-Wood)	1	0.7 g	2 min	Serving_Tray x1
Light green (e.g. PLA Matte Apple Green)	1	0.2 g	1 min	Plant_Leaves_03 x1
Silver silk (e.g. PLA Silk+ Silver)	1	0.2 g	1 min	Silver_Heart_Plant x1
Bone / ivory white	1	0.1 g	1 min	Pampas_Spray x1

Two-colour parts on a single-colour printer

In Bambu Studio's preview, drag the layer slider to the height below, right-click the **+** and add a **Pause** (or *Change filament*). When the printer stops, swap the filament and resume.

Part	Height	First colour	Second colour
`Coffee_Table` (printed upside down)	first layer **above 2.6 mm**	White (table top)	Light wood (apron and legs)
`Name_Plate` (printed on its back)	first layer **above 1.0 mm**	White (plate)	Dark brown/black (raised name)

5. Per-order customisation

1. **Name plate:** open `cad/parts/Name_Plate.scad` in OpenSCAD, change the `NAME = "...";` line, press F6, then File > Export > STL. Command line: `openscad -D 'NAME="Ananya"' -o Name_Plate.stl cad/parts/Name_Plate.scad`. The text auto-fits to 17 mm wide.
2. **Arch print:** make the artwork 72.6 × 173.3 mm (marble + gold veins + customer cut-out), using page 1 of `templates/Store_Versile_Print_Templates.pdf` as the guide.
3. **Photos:** 3 small portrait (22.6 × 26.4), 3 small landscape (25.6 × 23.4), 2 large portrait (28.6 × 38.4), plus the quote insert. Keep the image inside the blue window with a 1.4 mm white mat. Template page 2.
4. Print the PDF at **100 % / Actual size**. Measure the 50 mm test bar before cutting.

6. Assembly (three modules, then the box)

Positions are in mm. X is measured from the inside left wall, Y from the back wall forward, Z from the inside floor. Exact positions for every piece are on the *Assembly positions* sheets in `drawings/Store_Versile_Drawings.pdf`.

Prepare 1. Remove any brims and lightly sand the edges. Slide the photos into the frames from the top, with a dot of glue. 2. Stick the wood sticker on the three floor tiles. Cut the sticker open over the wall-panel slots at the back of tiles L and R. 3. Glue the arch print inside the arch's raised border. Glue the name plate on the print, centred, 150–158 mm above the arch bottom.

Module L (tile L, x 0–106) 1. Glue `Left_Wall_Panel_Lower` into the slot, flutes facing forward. Push `Left_Wall_Panel_Upper` onto its two key tabs and glue the seam. 2. Push 6 filament pins (1.75 mm × 6.4 mm) through the panel holes from the back, so 4 mm stick out. Slide the 3 `Floating_Shelf` parts onto the pins and glue. 3. Glue the large frames on the flutes: F4 at x 68, z 135; F5 at x 68, z 76 (bottom-left corners). 4. TV console: glue the white drawer plate into the front recess. Glue the 4 legs into the sockets, splaying outward (turn the right-hand pair 180°). Glue the console to the tile at x 3, touching the panel. 5. On the console: gramophone at x 10 (horn into its angled hole), small fern pot (`Plant_Pot_04` + `Plant_Leaves_04`) at x 44, 3 books at x 74. 6. Shelves, top to bottom: LOVE + frame; frame + leafy plant (`Plant_Pot_02`); succulent (`Plant_Pot_03`) + frame.

Module C (tile C, x 106–218) 1. Glue the arch standing on the tile at the back, centred at x 162. 2. Floor palms: push `Plant_Leaves_01A` into the front slot and `01B` into the rear slot of each `Plant_Pot_01`. Glue the pots at x 118 and x 206, 6 mm in front of the back wall. 3. Coffee table centred at x 162, 5 mm in front of the back wall. On it: tray, two mugs (handles outward), arch vase at the right end with the silver hearts pushed into its hole.

Module R (tile R, x 218–300) 1. Right wall panels: same as module L. 2. Glue `Right_Display_Shelf` on the tile against the panel, left board datum at x 223. 3. Fill the cells, top to bottom: frame | gold ornament (+ green leaves glued behind the eggs); cup | frame; donut vase + pampas | quote plaque | small fern; bunny | frame; 2 books | gold elephant.

Into the box 1. Join the three tiles with their dovetails. Dry-fit them in the box: the arch and panels touch the back board. 2. Glue the tiles to the box floor and the panels and arch to the back board (thin CA or double-sided tape). 3. Fit the LED strip under the top front edge (optional printed channel). Clean the glass, then close the frame.

7. Quality checklist (every order)

- [ ] Name spelled correctly and centred on the arch
- [ ] 8 photos, the right ones in the right frames, nothing crooked; the quote is in the plaque
- [ ] No glue strings or haze on the glass side
- [ ] All plants upright, leaves not touching the glass (≥ 3 mm gap)
- [ ] LED lights evenly; battery or cable is reachable
- [ ] Nothing rattles when the frame is gently tilted

8. First test prints (do these before production)

Test	Print	Pass when	If not, change in `versile_parts.scad`
Photo slot	`Photo_Frame_Small_P` + a cut photo-paper insert	Insert slides in fully, no gap at the window	slot (0.5) in `photo_frame()`
Panel key + shelf pin	Both wall panels (L) + 1 shelf	Panels align flute-to-flute; shelf slides onto 2 pins	`CLR` (0.2), `PIN_HOLE` (1.95)
Leaf fit	`Plant_Pot_04` + `Plant_Leaves_04`	Tab pushes in by hand and holds	`CLR`
Console legs	`TV_Console_Body` + 4 legs	Pegs press in, console stands level	leg socket (1.7)
Fine detail	`Plant_Leaves_01A`, `Pampas_Spray`, `Bunny` at 0.12 mm	Leaflets complete, no stringing	slow down, more cooling, or use a 0.2 mm nozzle

9. Files

Folder / file	Contents
`stl/`	One print-oriented STL per part
`cad/versile_parts.scad`	Parametric OpenSCAD library (all dimensions)
`cad/parts/*.scad`	One editable file per STL, already in print orientation
`cad/assembly.scad`, `assembly/Store_Versile_Assembly.3mf`	Complete assembled room
`drawings/Store_Versile_Drawings.pdf`	Layout, assembly positions, joint details and a dimensioned sheet for every part
`templates/Store_Versile_Print_Templates.pdf`	1:1 paper templates
`preview/renders/`	Rendered views
`layout/layout.py`	The approved layout (single source of truth)

STEP export isn't produced: the OpenSCAD (mesh) workflow cannot write true STEP solids. The 3MF and the parametric `.scad` source cover the assembly and editing needs.